

- 2.1 [5] <§2.2> For the following C statement, write the corresponding RISC-V assembly code. Assume that the C variables f, g, and h, have already been placed in registers x5, x6, and x7 respectively. Use a minimal number of RISC-V assembly instructions.

```
f = g + (h - 5);
```

```
addi x7, x7, -5
add x5, x6, x7
```

- 2.2 [5] <§2.2> Write a single C statement that corresponds to the two RISC-V assembly instructions below.

```
add f, g, h
add f, i, f
```

```
//Solution
f = i + g + h
```

- 2.3 [5] <§§2.2, 2.3> For the following C statement, write the corresponding RISC-V assembly code. Assume that the variables f, g, h, i, and j are assigned to registers x5, x6, x7, x28, and x29, respectively. Assume that the base address of the arrays A and B are in registers x10 and x11, respectively.

```
B[8] = A[i-j];

sub x28, x28, x29
slli x28, x28, 8
add x10, x10, x28

ld x5, 0(x10)
sd x5, 64(x11)
```

- 2.4 [10] <§§2.2, 2.3> For the RISC-V assembly instructions below, what is the corresponding C statement? Assume that the variables f, g, h, i, and j are assigned to registers x5, x6, x7, x28, and x29, respectively. Assume that the base address of the arrays A and B are in registers x10 and x11, respectively.

```
slli x30, x5, 3          // x30 = f*8
add x30, x10, x30        // x30 = &A[f]
```

```

slli x31, x6, 3      // x31 = g*8
add  x31, x11, x31   // x31 = &B[g]
ld   x5, 0(x30)      // f = A[f]

addi x12, x30, 8     // x12 = A[f] + 8
ld   x30, 0(x12)     // A[f] = A[f+1]
add  x30, x30, x5     // A[f+1] = A[f+1] + A[f]
sd   x30, 0(x31)     // B[g] = A[f+1] + A[f]

//Solution
B[g] = A[f+1] + A[f];

```

- 2.5 [5] <§2.3> Show how the value 0xabcdef12 would be arranged in memory of a little-endian and a big-endian machine. Assume the data are stored starting at address 0 and that the word size is 4 bytes.

```

little-endian:0x12efcdab
big-endian:0xabcdef12

```

- 2.6 [5] <§2.4> Translate 0xabcdef12 into decimal.

- 2.7 [5] <§§2.2, 2.3> Translate the following C code to RISC-V. Assume that the variables f, g, h, i, and j are assigned to registers x5, x6, x7, x28, and x29, respectively. Assume that the base address of the arrays A and B are in registers x10 and x11, respectively. Assume that the elements of the arrays A and B are 8-byte words:

```

B[8] = A[i] + A[j];

//Solution. Use x15 and x16 as temporary
registers.
slli x30, x29, 3      // x30 = j*8
add  x30, x10, x30     // x30 = &A[j]
ld   x15, 0(x30)      // x15 = A[j]
slli x31, x28, 3      // x31 = i*8
add  x31, x10, x31     // x31 = &A[i]
ld   x16, 0(x31)      // x16 = A[i]
add  x16, x16, x15
sd   x16, 64(x11)

```

- 2.8 [10] <§§2.2, 2.3> Translate the following RISC-V code to C. Assume that the variables f, g, h, i, and j are assigned to registers x5, x6, x7,

x28, and x29, respectively. Assume that the base address of the arrays A and B are in registers x10 and x11, respectively.

```
addi x30, x10, 8 //x30 = x10 + 8= A[1]
addi x31, x10, 0 //x31 = x10 + 0 = A[0]
sd x31, 0(x30)    //A[1] = A[0]
ld x30, 0(x30)    // A[1] = A[0]
add x5, x30, x31  // x5 = A[1]+A[0]
```

```
//Solution
f = A[1]+A[0]
```